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Free Riding on Product Returns to Drive Profits

Elisa Montaguti, Scott Neslin, and Sara Valentini

Abstract

Returning products has become standard practice for consumers and a significant "pain point" for retailers. The authors contend that returns can be harnessed to increase profits. Doing so requires retailers to manage the interweaving dynamics of product returns and purchases. The strategy is to comanage a virtuous cycle, whereby current returns increase future purchases, and a vicious cycle, whereby current returns increase future returns. Marketing executes the strategy by generating purchases, and the returns that follow impose a direct cost due to the vicious cycle. However, the virtuous cycle of returns can offset the vicious cycle, enabling the retailer to "free ride" on returns by optimizing its marketing. The approach follows the decision support system paradigm by combining a conceptual model, a statistical model, data, and optimization. A core construct is a stock variable tracking consumers' memory of return experiences, which drives both the virtuous and vicious cycles. The authors optimize marketing spend while accounting for return stock. The best results occur when dynamics are incorporated into both the statistical and optimization models. The results suggest that managers should avoid strict return policies aimed at eliminating returns. Instead, they should design policies that optimally balance the long-term benefits and costs of returns.

Keywords

product returns, returns carryover effect, dynamic optimization, marketing ROI, return stock, return policies, online retailing, marketing spend allocation

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Product returns are the bane of retail management. Retailers bemoan the deduction from revenues they thought were in the bank and the costs of processing returned merchandise. The problem is acute in online retailing, where return rates easily approach 30%–40%.

There is no doubt that returns saddle retailers with a short-term reduction in profits. However, research suggests that returns may benefit retailers in the long run by increasing future sales. Bower and Maxham (2012) find that customers who make returns increase spending in the long run if return shipping costs are free. In an extensive empirical analysis, Petersen and Kumar (2015) find a positive relationship between current returns and future sales. Chen (2021) develops a structural model to investigate the returns/sales relationship and replicates the positive relationship, and El Kihal et al. (2025) uncover this relationship in their research. In a meta-analysis, Janakiraman, Syrdal, and Freling (2016) find that lenient return policies unsurprisingly increase returns but also increase purchases to a greater extent. While the psychological mechanism behind this virtuous cycle has not been definitively identified and boundary conditions remain to be clarified, the evidence for a positive relationship between current returns and future purchases is compelling.

This evidence is provocative because it suggests that retailers have an opportunity to manage returns to maximize long-term profits. How can this be achieved? Our work suggests that firms can improve long-term profits by devising a dynamic marketing strategy that accounts for the relationship between product returns and future purchases. We refer to this as "free riding" on product returns, which requires optimizing marketing dynamically over time.

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The complexity arises from the dynamics among marketing, returns, and purchases. Marketing can produce a purchase in period t that may be returned in period t. Due to the positive impact of returns on purchase, this would increase the chance of a purchase in $t+1$—the virtuous cycle.[1] However, it may also increase the likelihood of a return in $t+1$—the vicious cycle. If the virtuous cycle is large relative to the vicious cycle, the retailer can free ride on the period t return and allow it to generate a purchase in period $t+1$. The retailer does not have to market in $t+1$; instead, it can market in $t+2$ to generate a purchase and repeat the sequence.

Marketing therefore plays a pivotal role because it can stimulate purchases that may be returned. These returns can generate future purchases due to the virtuous cycle as well as additional returns due to the vicious cycle. In this sense, marketing stimulates both cycles. The "trick" is to use marketing to generate enough returns to free ride on the virtuous cycle, but not too many returns, which would amplify the vicious cycle. This requires a statistical model that captures these dynamics and a dynamic optimization that increases long-term profits.

In this article we aim to show how retailers can use marketing to optimize long-term profits by managing returns. Our goal is to (1) develop a framework that describes the relationships among current returns, future returns, and purchases; the interplay between these vicious and virtuous cycles; and the role of marketing in these dynamics; (2) develop and estimate the statistical model that follows from this framework; (3) using those estimates as inputs, apply dynamic programming to generate an optimal free-riding policy; and (4) gain insights into how dynamic optimization increases profits. Does it do so by free riding on the future benefits of returns? How does this affect marketing spend, scheduling, and allocation across activities? How does the free-riding strategy interact with the firm's product return policy?

By combining a model, data, statistics, and optimization, our approach follows the decision support system (DSS) paradigm (Lilien, Kotler, and Moorthy 1992; Little 1979). The model is the framework:[2] It identifies key phenomena—virtuous and vicious cycles—and how they relate to marketing. The data consist of customer-level purchases and returns, while the statistical component comprises equations that mathematically represent the framework and the econometrics required to estimate them. These estimates serve as the descriptive tool at the core of the DSS. The optimization is a stochastic dynamic program.

A DSS produces tangible recommendations and insights that guide decision-making. For example, in our study, we recommend concrete marketing scheduling and allocation. One insight is that the goal is not zero returns but rather to use marketing to generate enough returns to free ride on the virtuous cycle without generating too much negative impact—the vicious cycle.

Our data cover a five-year period and include approximately 1,500 customers from several countries, observed from their first purchase and known to have made at least one purchase per year. The data are provided by an online fashion retailer with a "standard" 30% return rate. We show the role that customers' memories of return experience play in shaping virtuous and vicious cycles and measure the impact of marketing on these cycles. We also examine how optimal marketing varies across countries, which, due to different return policies, differ in their potential to generate profits through free riding.

The estimated statistical model identifies a positive direct impact of current returns on future sales—the virtuous cycle—and a positive direct impact of current returns on future returns—the vicious cycle. We show that dynamic optimization can prescribe a marketing policy that pulses spending to exploit the virtuous cycle without overgenerating the vicious cycle.

We also show that dynamic optimization outperforms myopic optimization that does not account for the future benefit of returns, and that the benefit of dynamic optimization is not realized unless the statistical model also incorporates dynamics. A successful free-riding strategy may therefore skew marketing spend toward activities that have a large impact on sales but a smaller direct impact on returns. For example, free (outbound) shipping is problematic because it has a substantial impact both on sales and returns. The dynamic optimization consequently reallocates marketing spend away from free shipping and toward advertising that builds sales without exerting a large direct effect on returns.

We further show that the firm's return policy influences the profitability of free riding. Strict policies reduce returns but in so doing also reduce the opportunity to free ride on them. Lenient policies can strike a better balance between the long-term benefits and costs of returns, thereby opening the door to free riding.[3] For instance, when our focal company implements strict return policies (e.g., in Denmark, Greece, and Russia), the difference between the benefits and costs of returns—between virtuous and vicious cycles—narrows, limiting the opportunity to capitalize on returns. In contrast, more lenient policies, as seen in the United States, increase this difference, offering greater opportunity to free ride on returns and leading to higher profits.

This article contributes in three ways: integrating existing knowledge, generating insights, and showing how to implement the free-riding strategy. First, we integrate a statistical model of virtuous and vicious cycles with a dynamic optimization to produce profitable free riding. To our knowledge, no previous research has done so. Second, in line with the DSS approach,

[1] To be concise, we sometimes use words like "influence," "impact," and "affect" that suggest causal relationships. Our results should be interpreted as associative and correlational and do not prove causality.

[2] The term "model" here is conceptual in the sense of Little (1979, p. 10), who defines it as "a preconceived idea of how the world works." We will use the term "statistical model" to refer to the equations that express this mathematically. In sum, we translate the conceptual model into a statistical model that we estimate with our data.

[3] We thank the anonymous reviewers for their insightful suggestions to examine regional differences, enabling us to distinguish regions with lenient versus strict return policies.

we offer insights into how to manage free riding. We show that lenient return policies are associated with higher levels of free riding and that free riding works not necessarily by reducing marketing spend but by carefully scheduling and allocating it across multiple marketing activities. We demonstrate that the evolution of consumers' memories of past return experiences, quantified by "return stock," is the fulcrum for generating virtuous and vicious cycles and thereby enabling successful free riding. We suggest that return stock translates into virtuous and vicious cycles by enhancing consumers' perceptions of fairness and their relationship with the company while reducing their concerns about purchase risk (buying the wrong product).

Also in line with the DSS paradigm, we investigate the key requirement for implementing the free-riding strategy: incorporating return dynamics into both the statistical and optimization models. Our formulation of return stock, combined with dynamic stochastic optimization, generates the largest free-riding profits. We find that successful implementation requires a balance between virtuous and vicious cycles; the objective is a profitable level of returns, not zero returns.

In summary, we show how a novel dynamic statistical model, optimized over time, can be applied in real-world contexts to free ride on the long-term benefits of product returns. By combining model, statistics, data, and optimization, our approach functions as a DSS. Free riding requires dynamics in both the statistical model and the optimization. Dynamic optimization paired with a nondynamic statistical model fails to capture the necessary long-term effects; likewise, a dynamic statistical model with myopic optimization fails to fully leverage those effects. A fully dynamic approach therefore provides a practical DSS for optimizing return management. By striking a balance that avoids amplifying the vicious cycle while still exploiting the benefits of returns, firms can achieve both financial and environmental gains.

Literature Review

We summarize research in marketing focusing on three streams relevant to our work: the impact of returns on future purchase (the virtuous cycle), the impact of returns on future returns (the vicious cycle), and the impact of marketing on returns.

The Virtuous Cycle

Emerging evidence suggests that returns can increase future purchases and revenues (Bower and Maxham 2012; Chen 2021; Petersen and Kumar 2009, 2015). This is intriguing and apparently not well accounted for by practitioners and academics. Previous research suggests two possible explanations for this virtuous cycle: risk reduction (Chen 2021; Petersen and Kumar 2015) and fairness (Bower and Maxham 2012).

Consumers often risk dissatisfaction by buying the "wrong" product. Anderson, Hansen, and Simester (2009) conceptualize returns as an economic option that enables consumers to remedy this risk. This makes clear that the ability to return encourages a customer to purchase now, but why would the return option also increase purchasing in the future?

One possibility is that product returns reduce consumers' future risk. Petersen and Kumar (2009) posit that returns enable consumers to learn more about a retailer's products, thus reducing the risk of buying the wrong product in the future. Venkatesan and Kumar (2004) argue that returns provide firms with the opportunity to increase customer satisfaction. However, too many returns may prevent the firm from taking advantage of this opportunity. Venkatesan and Kumar (2004) and Petersen and Kumar (2009) find an inverted U-shaped relationship between returns and future purchases.

Bower and Maxham (2012) show that when customers perceive that a retailer handles returns fairly, returning a product produces less regret and consumers may even experience "rejoicing or elation" (p. 133). This in turn increases customer satisfaction and generates more purchases in the future. If consumers perceive returns to be unfairly managed, they regret the purchase, presumably becoming dissatisfied and decreasing future purchases.

In sum, research to date suggests that risk reduction and perceived fairness play a role in the empirical finding that returns generate future purchases. More work is needed to probe these explanations as well as others. The key takeaway, however, is that returns can exert a positive impact on future purchases.

Petersen and Kumar (2015) leverage this finding to develop a customer lifetime value model accounting for product returns. They estimate this model and optimize marketing spend one period at a time. This "myopic optimization" indicates that retailers can increase profits by accounting for the benefits and costs of returns. Although their empirical model includes dynamics, they do not optimize dynamically. We propose that this is a key desideratum. An optimal marketing strategy might tolerate an acceptable level of returns in the short run to free ride on the virtuous cycle. Doing so requires the optimization to look ahead and incorporate the impact of current actions on future profits, which in turn requires both the statistical and optimization models to be dynamic.

The Vicious Cycle

El Kihal et al. (2025) investigate how return rates evolve over time. Returns provide customers with an opportunity to learn about the brand's product quality and fit, potentially decreasing future returns. Return rates, however, could increase because customers develop a return habit or become more discerning over time. The authors find that return rates increase over time, suggesting a "vicious cycle" whereby current returns beget future returns. This finding is important as it demonstrates that the direct impact of current returns on future returns must be included in any empirical model.

The Impact of Marketing on returns

Shehu, Papies, and Neslin (2020) suggest that free-shipping promotions act as a risk premium that encourages riskier

purchases and a commensurate increase in returns. El Kihal and Shehu (2022) posit that marketing elevates both the expected and experienced value of a purchased product. If the expected value is less than the experienced value, the product is likely to be returned. The authors find that several marketing instruments increase returns, suggesting that marketing should be included when modeling the interplay among marketing, purchases, and returns.

Table 1 summarizes the preceding discussion and highlights what distinguishes this article. Our work is unique in integrating a dynamic descriptive model with a dynamic optimization of marketing. To our knowledge, no prior research has combined these approaches. This integration is essential because, as we show, neglecting one of these dynamics undermines the other.

In addition, our work is uniquely comprehensive. Specifically, it investigates mediators that serve as mechanisms for translating returns into future behavior, considers the impact of multiple marketing activities on returns, conceptualizes return stock to quantify consumers' memory of the return experience, and empirically studies the impact of different return policies. While previous studies examine a subset of these issues, none considers them jointly.

Our work can thus be viewed as a DSS (Little 1979), as it combines a model, statistics, data, and optimization to generate tangible marketing plans and conceptual insights for free-riding returns. For example, we show that the gap between the magnitudes of virtuous and vicious cycles provides room for free riding: Strict return policies minimize this gap, whereas lenient policies increase it. Lenient policies therefore provide the best opportunity for free riding. Capturing these dynamics is important, yet to our knowledge no prior work has investigated them in this integrated manner. Our approach explicitly models these inherently dynamic cycles and optimizes decisions accordingly.

Conceptual Framework

Figure 1 portrays the conceptual framework suggested by the previous discussion, depicting the relationships among returns, marketing, and customer-level sales. The figure shows that marketing induces both sales and returns, as documented in our literature review, and that the role of returns is captured by return stock. The Engel, Blackwell, and Miniard model of consumer decision-making (1995, p. 152) motivates this concept. Because return stock is a key component of our framework, we elaborate on it subsequently.

The Engel, Blackwell, and Miniard model posits that memory governs beliefs, attitudes, and intentions regarding future behavior. In our framework, return stock captures this memory. We are concerned with two behaviors: returns and purchases. When considering whether to return a product, customers assess their beliefs about factors such as ease of returning and hassle costs (Bower and Maxham 2012). If, based on memory, these beliefs are favorable, they foster attitudes and intentions that encourage returns, creating the vicious cycle. However, memory-supported beliefs that the retailer handles returns fairly and expeditiously create favorable attitudes toward the retailer, increasing future purchases. This is the virtuous cycle.

The following factors help create memory: First, memory is strengthened by multiple experiences. Second, the more recent these experiences, the stronger the memory. Third, a memory not strengthened by multiple recent experiences fades away. For instance, if a customer returns four items within three months, they will have a strong memory of returning products. Conversely, if it has been three years since the last return, the experience is likely forgotten. The process of building memory through repeated recent experiences is "wear-in," whereas losing it is "wearout" or "forgetting" (Engel and Blackwell 1982, pp. 249, 483).

We thus operationalize return stock as a "stock variable," consistent with previous advertising literature (e.g., Danaher, Bonfrer, and Dhar 2008; Dinner, Van Heerde, and Neslin 2014) and the return habit variable used by El Kihal et al. (2025). Return stock equals previous return stock multiplied by a decay parameter to represent forgetting, plus a recent-experience factor incorporating recent returns. Forgetting produces wearout, while recent experience produces wear-in. A high propensity to forget coupled with only sporadic experiences yields a smaller return stock.

In sum, return stock reflects the connection between past behavior and future actions, making it central to understanding virtuous and vicious cycles. Marketing can set these dynamics in motion through its direct impact on sales or returns. This suggests that marketing can be adjusted—in terms of spending, scheduling, and allocation across activities—to leverage returns to drive sales. Such adjustments increase profits if the gains triggered by the virtuous cycle exceed the costs associated with vicious cycles. A crucial issue for marketing, therefore, is the strength of its direct effects on sales and returns and its indirect effects through return stock. We measure these using a dynamic statistical model to capture return stock and a dynamic optimization to recommend marketing spend, timing, and allocation among activities.

Research Context and Data

Our research context is online retailing in the fashion industry (e.g., clothing, shoes, accessories). This is a particularly appropriate setting for our study for several reasons. First, online sales of fashion items represent more than one-third of e-commerce revenues worldwide. They accounted for $668 billion (U.S. dollars) in 2021 (Statista 2022a), underscoring the importance of this industry to the global economy.

Second, returns data (Statista 2022b) show that clothing and fashion items have significant return rates: 26% in the United States, 32% in the United Kingdom, and even higher in some European countries (e.g., Austria, Germany, and the Netherlands at 33%). The business press documents online retailers trying to find an optimal way to manage product returns (Ader et al. 2021; Eley 2022). Many retailers (e.g., Asos, Zara, Net-a-Porter) monitor competitors' return policies and consumers' return

Table 1. Literature Review Summary.

Research	Returns → Future Purchase (Virtuous Cycle)	Returns → Future Returns (Vicious Cycle)	Mediating the Virtuous and Vicious Cycles	Number of Marketing Activities Influencing Returns	Quantifying the Returns Experience	Return Policy	Dynamic Descriptive Model	Dynamic Optimization of Marketing	Dynamic Model and Dynamic Optimization
Petersen and Kumar (2009)	✓	✓		Single	Total returns	Single	Yes	Myopic	No
Petersen and Kumar (2015)	✓	✓		Single	One-period lag	Single	Yes	Myopic	No
Chen (2021)	✓	✓		Single		Single	Yes		
El Kihal et al. (2025)	✓	✓		Multiple	Stock variable		Yes		
El Kihal and Shehu (2022)				Multiple					
Shehu, Papies, and Neslin (2020)				Single			Yes		
Wood (2001)	✓	✓				Strict vs. lenient			
Bower and Maxham (2012)	✓		Fairness			Free vs. fee			
This article	✓	✓	Fairness, relationship, risk	Multiple	Stock variable	Strict vs. lenient	Yes +	Yes →	Free riding

Notes: Empty cells indicate aspects not addressed by the respective study.

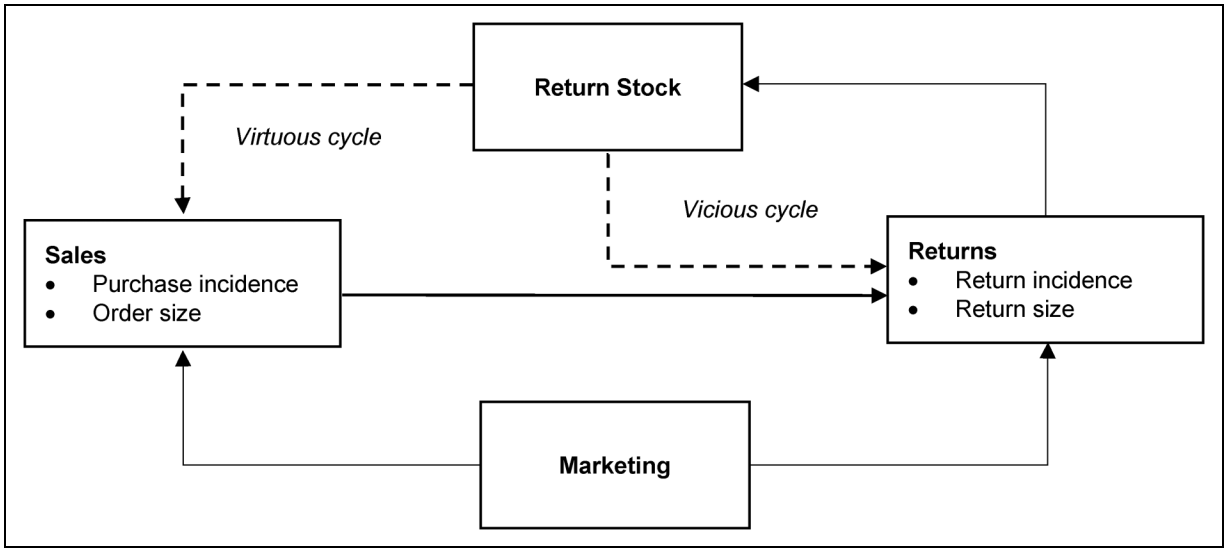


Figure 1. Conceptual Framework.

behavior (Eley 2022).[4] Perhaps overlooking the virtuous cycle, some charge for returns (e.g., Boohoo) or make returns more onerous for customers (e.g., Amazon). Returns are therefore highly relevant to fashion industry management.

Third, fashion retailers spend substantial amounts on price promotions and marketing communications to boost sales. Given its role in influencing both vicious and virtuous cycles, marketing serves as the decision variable in our optimization.

Data

We obtained data from a leading retailer that sells fashion items online. Its core products are clothing and shoes for women, men, and kids, and it also sells accessories, home décor, and art. It carries a large assortment of brands as well as its private label and operates worldwide.

Our framework requires data on the incidence and monetary value of purchases and returns. When analyzing dynamics, it is useful to observe consumers from their first purchase to avoid confounding dynamic and cross-section effects. We also need detailed information about the firm's marketing efforts over a relatively long observation period. The retailer provided us with data and information that fulfilled these requirements.

We obtained a random sample of new customers acquired in 2013, 2014, or 2015. Each customer is observed from their first purchase until the end of 2017. We also collected demographics (age, gender, country of residence, country of purchase). Additionally, the firm provided its marketing calendar for each country and day over the 2013–2017 period. We organized the information in the marketing calendar to create four marketing variables: retailer advertising, brand advertising, markdowns, and free shipping.

We focus on active customers, defined as those who purchased at least once each year after they were acquired. We used this sample for several reasons. First, these customers are managerially relevant: Although they represent a minority of customers, they account for 61% of revenues (Web Appendix A, Table WA.2). Second, the requirement avoids survivor bias. Without the "every-year" requirement, we would have included customers who "churned." Customers remaining at the end of our data would be "survivors," different from those we started with. Thus, what we infer as dynamic behavior could instead have reflected cross-customer differences.[5] Third, focusing on longer purchase histories increases estimation precision. Intuitively, a longer purchase series provides more within-customer information for detecting dynamics. Theoretically, work related to initial conditions suggests that longer time series improves estimates of dynamics (Keane 1997; Simonov et al. 2020). Finally, this approach is consistent with previous research, which constructs long time series to estimate dynamics (Ansari, Mela, and Neslin 2008; Dubé, Hitsch, and Rossi 2010; Erdem and Keane 1996; Erdem, Imai, and Keane 2003; Mark et al. 2013; Seetharaman 2004).

The final sample includes 1,498 customers and 291,096 observations, analyzed at the weekly level.[6] Customers live in 78 different countries (52% in Europe, 16% in Russia, 15% in the United States and Canada, 11% in Asia, and 6% in the rest of the world). The return rate, as measured by the percentage of purchases for which at least one item is returned, is 33%. Figure 2 shows that return rates differ by country, and Figure 3 shows how they trended over time.

[4] Our data lack competitive information, limiting our analysis in this respect, but our region-specific findings suggest that firms should monitor returns other than their own.

[5] We thank the Associate Editor for valuable insights on this issue. See Web Appendix A for a discussion suggesting that survivor bias might not have been crucial in our application. However, at the outset, we thought it prudent to guard against it.

[6] There were 330 customers acquired in 2013 observed for five years (85,800 obs.), 444 acquired in 2014 observed for four years (92,352 obs.), and 724 acquired in 2015 observed for three years (112,944 obs.).

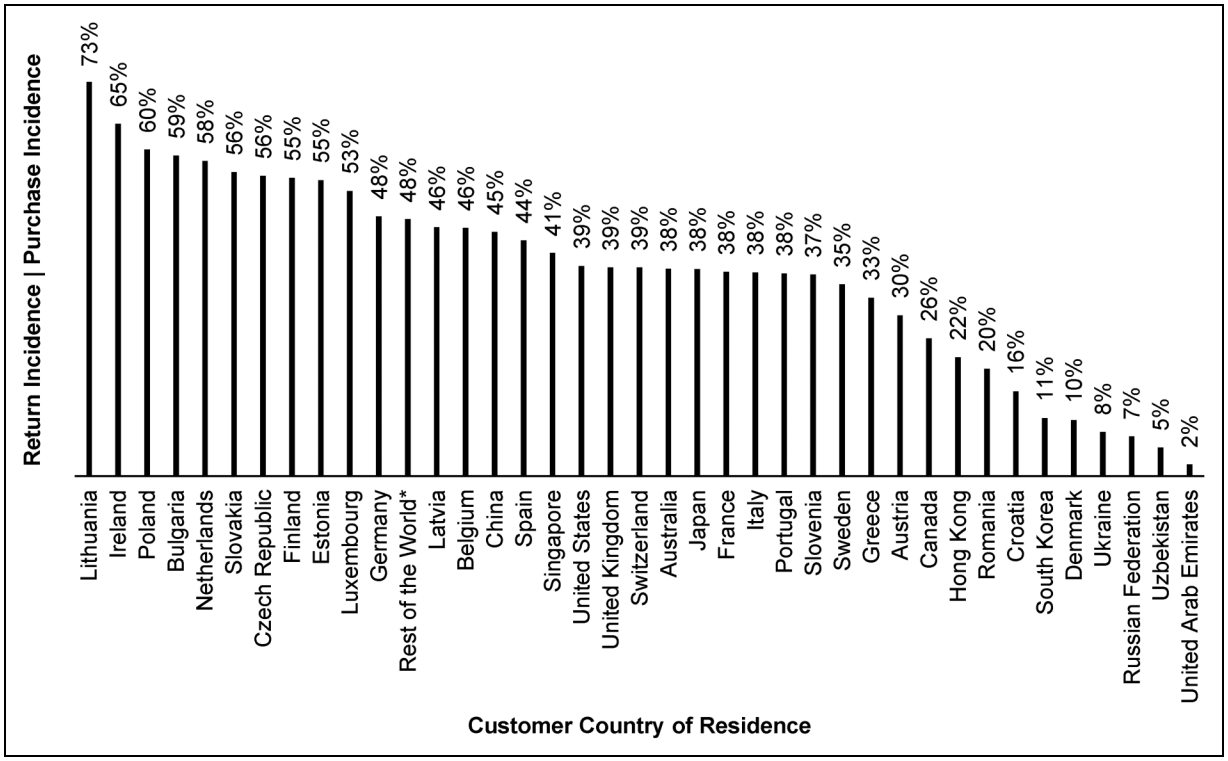


Figure 2. Return Rate by Country of Residence.
Notes: The y-axis reports return incidence conditional on purchase (i.e., the percentage of purchases in which at least one item is returned).

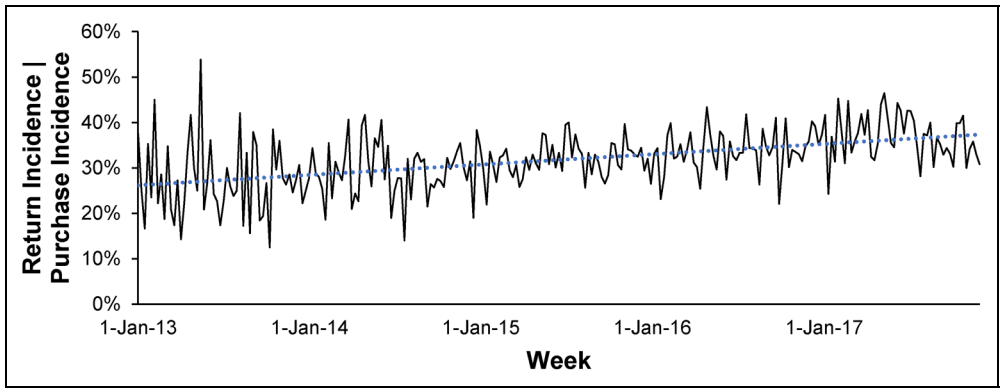


Figure 3. Return Rate over Time.
Notes: The y-axis reports return incidence conditional on purchase (i.e., the percentage of purchases in which at least one item is returned).

Table 2 provides descriptive statistics for the outcome variables of interest: purchase incidence; order size (€), net of return size; return incidence (whether the customer returned at least one item from a given week's purchase); return size (€), defined as the value of the items returned; and customer profit per week.[7] In 2013, customers returned at least one item from 26.54% of their purchases, at an average value of €167.21 per return.

Marketing Activity

The retailer's core marketing activity varies by country and week but not by customer—it did not target marketing at the customer level. The retailer also did not advertise on television, print, or radio, instead reaching consumers through display advertising on its website partners' websites, through email, and/or through social media posts.

Using documents provided by the retailer, we delineated four types of marketing campaigns: markdown (MKD), free shipping (FS), retailer advertising (RET), and brand advertising (BRD). Markdown campaigns offered a range of discounts, with reductions between 40% and 90%, typically at the end of

[7] In total, 33% of customers who purchased in a given year returned at least one item. The average return size is €185, whereas the average order size is €210. This means that customers typically returned part of the order.

Table 2. Descriptive Statistics of Purchase and Return Behavior.

	Purchase Incidence (1 = Yes, 0 = No)		Order Size per Purchase (Net of Return Size)		Return Incidence | Purchase (1 = Yes, 0 = No)		Return Size per Return		Profits per Customer per Week[a]	
Year	Percentage	SD	Mean (€)	SD	Percentage	SD	Mean (€)	SD	Mean (€)	SD
2013	8.06	27.22	224.43	216.70	26.54	44.18	167.21	162.60	2.08	45.17
2014	9.00	28.62	205.55	216.99	28.90	45.34	168.37	181.18	5.64	56.81
2015	9.08	28.73	214.12	264.17	32.22	46.74	187.86	243.91	15.18	86.75
2016	10.77	31.00	208.41	265.12	33.70	47.27	186.55	255.92	16.70	92.02
2017	10.25	30.33	209.99	274.79	36.02	48.01	190.50	269.37	15.67	90.51
Total	9.77	29.70	210.69	259.90	33.23	47.11	185.26	247.04	11.05	76.99

[a]Profit calculations follow the methodology described in Equation 8, incorporating actual values of marketing and customer behavior within the sample during the observation period.

the season. Free-shipping campaigns did not follow a consistent timing pattern and were not associated with specific brands; rather, they applied to the entire order. Retailer advertising aimed to boost the retailer's brand and promote a specific feature (e.g., wish list, private label, sustainable actions room). These campaigns often included shallow discounts (10% or 15% off), typically co-occurring with holidays or special events (e.g., Mother's Day, preview of the new season). Brand advertising featured manufacturer brands carried by the retailer and did not involve discounts. The retailer funded brand advertising campaigns itself, without co-participation from manufacturers. This reflects the role of the focal firm as an outlet e-retailer that manages its brand assortment and related marketing actions autonomously. Figure 4 provides examples of these campaign types.

Operationalization of the Marketing Variables

We know the start and end dates of each campaign in each country. Therefore, for each week t and country c, we observe the number of days the retailer ran each campaign type and operationalize the marketing variables accordingly (for a similar approach, see Blattberg and Neslin [1990, p. 368] and Mulhern and Leone [1991, p. 70]). For example, let week t be the week of Monday, February 6, through Sunday, February 12, and let c be the United States. If the retailer ran a retailer advertising campaign in the United States from February 6 to February 8 and a free-shipping campaign from February 7 to February 10, retailer advertising (RET) would equal three days in week t, country c (United States), and free shipping (FS) would equal four days. Table 3 summarizes the marketing variables.

Because marketing is operationalized as the number of days each campaign type ran in week t and country c, the minimum value of each marketing variable is 0. The maximum may be greater than 7, as the retailer could run two or more campaigns of the same type in the same week. For example, the retailer might run two brand-building campaigns, each for a different brand and each lasting five days, in week t. In that case, BRD_t would equal 10 days. For instance, in 2017 the retailer offered markdowns (MKD) on average .30 days per week, with a maximum of 7, and retailer advertising (RET) on average .65 times per week, with a maximum of 8.

Statistical Model

The DSS conceptual model—our framework in Figure 1—suggests four customer decisions: (1) whether to purchase in a given week (purchase incidence); (2) how much to spend, conditional on purchase (order size); (3) whether to return at least one item from a given week's purchase (return incidence); and (4) how much to return, conditional on returning (return size). Accordingly, we specify a statistical model consisting of four equations. Purchase incidence is conditional on positive latent utility for purchase, and order size is observed only if a purchase occurs. Return incidence is conditional on purchase and positive latent utility for return. Return size is observed only if both purchase and return occur. Purchase and return incidence are estimated using probit models, whereas order and return size are estimated using regressions in which the dependent variables are log-transformed. The four equations are the following:

Purchase

$$P_{it} = \begin{cases} \text{Purchase} & \text{if} \quad P_{it}^{*} > 0, \\ \text{No Purchase} & \text{if} \quad \text{otherwise}, \end{cases} \tag{1.a}$$

where

$$P_{it}^{*} = \alpha_i^{P} + \gamma^{P}\text{ReturnStock}_{it} + \beta_M^{P}\text{Mktg}_{ct} + \beta_{MLag}^{P}\text{Mktg}_{ct-1} + \iota^{P}P_{it-1} + \kappa^{P}CC_i + \nu^{P}T_t + u_{it}^{P}. \tag{1.b}$$

Order Size

$$OS_{it} = \alpha_i^{OS} + \gamma^{OS}\text{ReturnStock}_{it} + \beta_M^{OS}\text{Mktg}_{ct} + \beta_{MLag}^{OS}\text{Mktg}_{ct-1} + \iota^{OS}OS_{it-1} + \kappa^{OS}CC_i + \nu^{OS}T_t + u_{it}^{OS} \quad \text{if Purchase; else unobserved.} \tag{2}$$

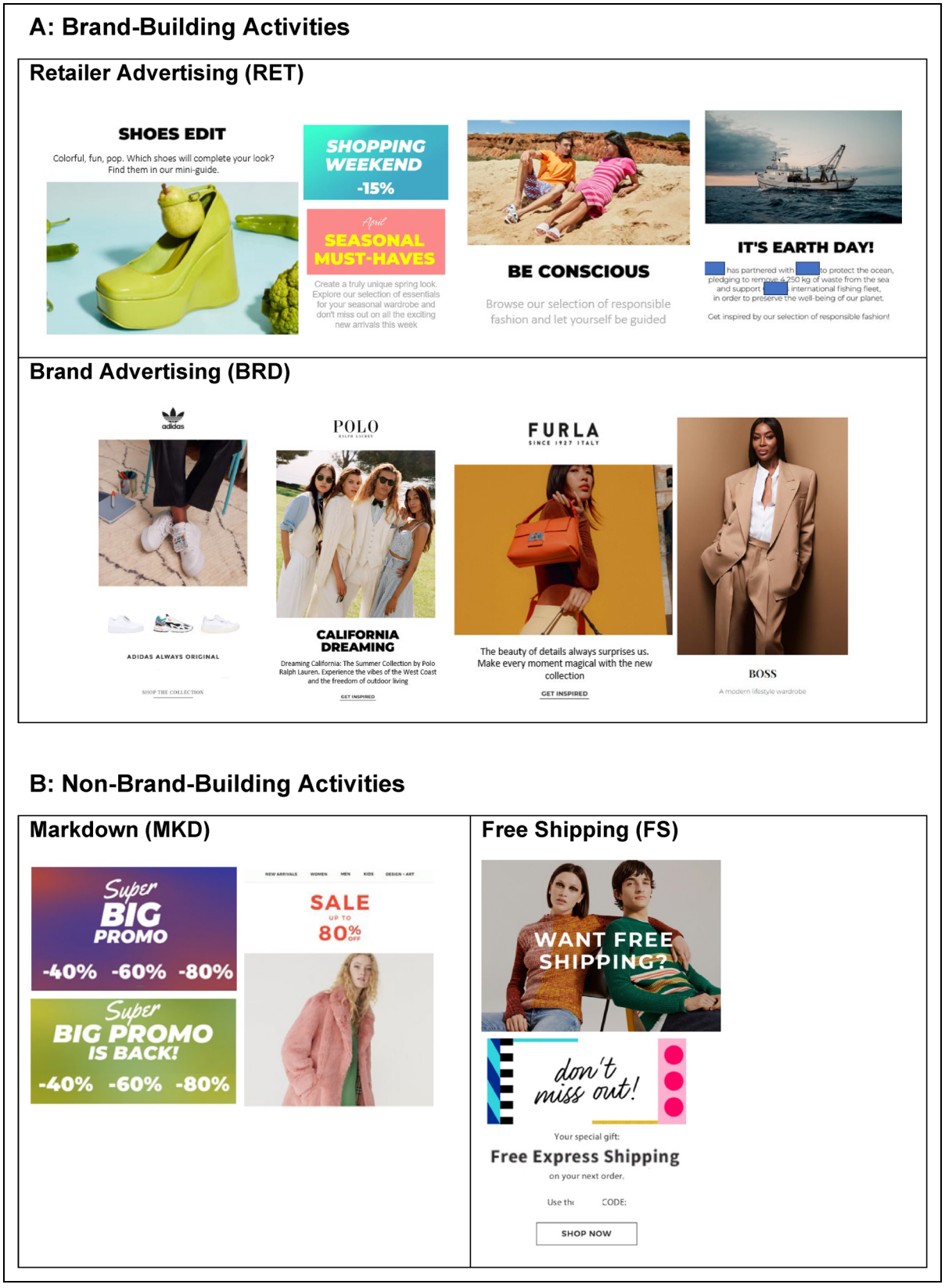


Figure 4. Examples of Marketing Activities.
Notes: The retailer's name has been removed from selected examples. Brand names shown in the BRD section are used solely for illustrative purposes and do not reflect any specific partnership.

Table 3. Frequency of Marketing Activity.

A: Non-Brand-Building Activity

	Discounts (MKD)[d]						Free Shipping (FS)					
Year	M[a]	SD	% Nonzero[b,c]	Cond. M[d]	Min	Max	M[a]	SD	% Nonzero[b]	Cond. M[d]	Min	Max
2013	.47	.62	40.47	1.17	0	2	.07	.43	3.00	2.34	0	3
2014	.92	1.82	41.88	2.20	0	7	.08	.45	3.28	2.44	0	3
2015	.35	.60	30.88	1.14	0	5	.09	.46	3.29	2.73	0	3
2016	.33	.57	28.83	1.13	0	4	.08	.45	3.24	2.47	0	3
2017	.30	.47	30.08	1.01	0	7	.10	.47	3.54	2.82	0	3
Total	.48	.82	34.43	1.33	0	7	.09	.46	3.21	2.80	0	3

B: Brand-Building Activity

	Manufacturer Brand (BRD)						Retailer Brand (RET)					
Year	M[a]	SD	% Nonzero[b]	Cond. M[d]	Min	Max	M[a]	SD	% Nonzero[b]	Cond. M[d]	Min	Max
2013	.53	.60	47.61	1.12	0	2	1.37	2.28	45.57	3.00	0	7
2014	3.43	2.97	67.53	5.08	0	8	.65	1.84	12.66	5.13	0	8
2015	1.87	2.34	50.33	3.71	0	7	.64	1.83	12.43	5.13	0	8
2016	3.37	1.82	90.69	3.72	0	7	.64	1.83	12.41	5.16	0	8
2017	3.38	1.70	92.12	3.67	0	7	.65	1.85	12.58	5.17	0	8
Total	2.52	1.88	69.66	3.46	0	8	.79	1.93	19.13	4.72	0	8

[a]M is the mean number of days per week each marketing activity was implemented, across countries.
[b]"% Nonzero" indicates the percentage of weeks with nonzero marketing.
[c]The percentage discount for MKD varies with deep discounts over 70%, usually at the end of the season, occurring in 8.17% of cases.
[d]"Cond. M" indicates the mean conditional on the presence of the marketing campaign.

Return Incidence

$$R_{it} = \begin{cases} \text{Return} & \text{if} \quad R_{it}^{*} > 0, \\ \text{No Return} & \text{if} \quad \text{otherwise}, \end{cases} \tag{3.a}$$

where

$$R_{it}^{*} = \alpha_i^{R} + \gamma^{R}\text{ReturnStock}_{it} + \beta_M^{R}\text{Mktg}_{ct} + \beta_{MLag}^{R}\text{Mktg}_{ct-1} + \tau^{R}\text{OS}_{it} + \kappa^{R}\text{CC}_i + \nu^{R}\text{T}_t + u_{it}^{R} \quad \text{if Purchase; else unobserved} \tag{3.b}$$

Return Size

$$\text{RS}_{it} = \alpha_i^{RS} + \gamma^{RS}\text{ReturnStock}_{it} + \beta_M^{RS}\text{Mktg}_{ct} + \beta_{MLag}^{RS}\text{Mktg}_{ct-1} + \iota^{RS}\text{RS}_{it-1} + \tau^{RS}\text{OS}_{it} + \kappa^{RS}\text{CC}_i + \nu^{RS}\text{T}_t + u_{it}^{RS} \quad \text{if Purchase and Return; else unobserved.} \tag{4}$$

As suggested by our DSS-motivated conceptual model (Figure 1), ReturnStock_{it} represents consumer i's memory or past return experiences, drawing on the concept of "adstock" (Broadbent 1984; Danaher, Bonfrer, and Dhar 2008). Define $\text{ReturnShare}_{it-1}$ as the ratio of return size (RS_{it-1}) to order size (OS_{it-1})—that is, the fraction of an order that is returned. The parameter λ parsimoniously captures wear-in and forgetting. A large λ increases wear-in and inhibits forgetting. Thus, a vivid experience boosts memory significantly and is less likely to be forgotten easily. Using return share is consistent with the theory of interference in memory research, which suggests that returning only part of an order leaves a weaker memory trace than returning the entire order because competing information interferes with memory encoding and retrieval (Kliegl and Bäuml 2021). ReturnStock is then

$$\text{ReturnStock}_{it} = \lambda\,\text{ReturnStock}_{it-1} + \text{Return Share}_{it-1}\,. \tag{5}$$

The parameters γ^{P} and γ^{OS} in Equations 1 and 2 quantify the virtuous cycle, capturing the impact of ReturnStock on purchase. The parameters γ^{R} and γ^{RS} in Equations 3 and 4 quantify the vicious cycle, quantifying the impact of ReturnStock on returns. Per our framework, we include the direct impact of marketing on both purchase and returns by including Mktg_{ct} in all four equations. Mktg_{ct} is a four-element vector of the four marketing types: retailer brand advertising, brand advertising, free shipping, and markdown. The subscripts c and t indicate that marketing activity varies by country and week.

We include several control variables. Lagged values of the dependent variable in Equations 1, 2, and 4 (ReturnStock already includes lagged returns, so no additional lag is included in Equation 3) reflect carryover in the dependent variable separate from other dynamics, notably ReturnStock_{it}. CC_i are customer i's descriptors and include gender, age, country of residence, and acquisition cohort, as well as past behavior measured in the 26 weeks before the beginning of the observation window. These include the number of purchases, average

order size, return share, average return size conditional on return, and the proportion of purchases made in each channel (tablet, smartphone, or desktop).[8] Additional details are provided in Web Appendix A.

The four equations also control for time trends (T_t), using fixed effects for month and year of purchase. Finally, we include current order size OS_{it} in Equations 3 and 4, reflecting the assumption that larger orders are more likely to result in at least one item being returned and, naturally, in larger monetary return amounts.

The error terms follow a multivariate normal distribution $u_{it}^{P}, u_{it}^{OS}, u_{it}^{R}, u_{it}^{RS} \sim N(0, \Sigma)$. The covariance matrix Σ among the error terms is

$$\Sigma = \begin{bmatrix} 1 & \sigma_{OS,\,P} & \sigma_{R,P} & \sigma_{P,RS} \\ \sigma_{P,OS} & \sigma_{OS}^2 & \sigma_{R,OS} & \sigma_{OS,\,RS} \\ \sigma_{P,R} & \sigma_{OS,R} & 1 & \sigma_{R,RS} \\ \sigma_{P,RS} & \sigma_{OS,RS} & \sigma_{R,RS} & \sigma_{RS}^2 \end{bmatrix}. \quad (6)$$

As is common, we set the variance of the purchase and return probit model equations to 1.

We identify the parameters for OS_{it} in Equations 3 and 4 using exclusion restrictions. The CC_i variables differ across equations: Equation 1 includes total purchase incidence, Equation 2 includes average order size, and Equations 3 and 4 do not include these variables. Lagged order size OS_{it-1} should influence current order size OS_{it} but should not directly affect current return incidence R_{it} or return size RS_{it}. Therefore, OS_{it-1} appears in Equations 1 and 2 but not in Equations 3 or 4. The lagged dependent variable also differs across equations.

We leverage the customer-level panel data to identify the impact of ReturnStock. Recall that our sample includes customers who made purchases each year beginning with their acquisition year. This contributes to identification in two ways. The first is temporal variation: $ReturnStock_{it}$ captures customer i's return experience up to time t. Because subsequent behavior is observed after t, this reveals the impact of prior returns on future behavior, on a within-customer basis.[9] We also include customer-specific temporal controls such as lagged return and purchases. The second factor is theoretical evidence. Consistent with intuition, econometric work on initial conditions for data required by dynamic models suggests that more complete time series improve the identification of dynamic coefficients (Erdem, Imai, and Keane 2003; Simonov et al. 2020). As Simonov et al. (2020) note, "only when we use larger values of T^{purch} [do] we see the true values [of a loyalty effect]" (p. 797).[10]

These factors require sufficient within-customer variation over time in return incidence and return size, which the data provide. For example, only 26% of customers never returned a purchase, and 2% returned all their purchases. Thus, 72% returned at least once but not always. Among these, the mean return rate was 37% per customer, with a maximum of 93% and a minimum of 4%. Regarding return size, among customers who returned at least twice, 0% returned the same amount for each return. The mean return size per customer was €153, with a maximum of €1,407 and a min of €17. These return rate and return size numbers suggest adequate within-customer variation to measure the impact of returns.

A third factor aiding identification is cross-country variation. Countries vary in return policies—strict versus lenient—and in the timing of marketing, creating temporal variation. Thus, cross-country differences systematically generate both temporal and cross-sectional variation.

The preceding three factors support our claim of parameter identification, and the data provide substantial variation for identifying dynamics. However, we cannot guarantee causal inference. For example, we cannot capture how consumers' valuations of returns evolve over time, nor do we observe their opinions of the items they receive; these unobservables may influence subsequent purchase and return behavior. This limitation is consistent with the DSS viewpoint, which emphasizes insights as well as concrete policy. The resulting policies may not be exact because parameters are not quantified perfectly; however, the insights, while judgmental, remain.

We estimate the model jointly using the simulated maximum likelihood seemingly unrelated regression procedure in Stata's CMP package (Roodman 2011, p. 160), which constitutes full-information maximum likelihood. This approach considers the joint likelihood function of all equations, enabling consistent estimation (Lahiri and Schmidt 1978; Roodman 2011).

Model Optimization

A key step in the DSS paradigm is optimization—in our case, prescribing the marketing policy that exploits the virtuous and vicious cycles produced by product returns. The estimated statistical model provides the results required for this step. Specifically, we derive the four-element vector $Mktg_t$, representing the number of days in week t allocated to each marketing type: discounts (MKD), free shipping (FS), retailer advertising (RET), and brand advertising (BRD). Each campaign has its own parameter in Equations 1–4. We optimize over N customers for a finite horizon of $W = 25$ weeks. We also initialize $ReturnStock_{it}$ at .75;[11] the insights are not sensitive to the initial value. The optimization is

$$\max_{Mktg_t} \sum_{i=1}^{N} \sum_{t=1}^{W} Profit_{it}, \quad (7)$$

[8] This is for initialization. The first 26 weeks are not included in the estimation, yielding 252,148 observations instead of the 291,096 available.

[9] This also enables us to validate our results using propensity score matching (PSM; see Web Appendix F for more details). The close alignment between PSM and model-based results provides further support for our results.

[10] The authors gratefully acknowledge Asim Ansari (Columbia University) for his insights and suggestions on this issue.

[11] We set the initial value of ReturnStock via an iterative search procedure. Using a 25-week horizon, we explored initial levels from .45 to .75 (step .10) and chose the smallest value that yielded stable process behavior (control chart) trajectories for ReturnStock; .75 was the first to satisfy this criterion.

$$\begin{aligned}\text{Profit}_{it} = {} & P_{it} \times \text{Margin} \\ & \times \{OS_{it} - R_{it} \times [RS_{it} + \text{ProcessingCosts}_t]\} \\ & - \text{MKD_Spend}_t + \text{FS_Spend}_t + \text{RET_Spend}_t \\ & + \text{BRD_Spend}_t.\end{aligned} \quad (8)$$

Profit_{it} denotes profit for customer i, and P_{it} is the probability that customer i purchases in week t. We use the average margin provided by the retailer (40%), which is multiplied by revenue. Revenue depends on order size and return costs: Order size (OS) includes any price discount or free-shipping cost, and return costs equal the probability of return (R) times the return amount (RS), plus processing costs. Return size is customer-specific and predicted by our model; however, processing cost is the same for each customer and therefore subscripted by t. Management indicated that return costs were the same each week, €8, reflecting processing and return shipping.

Weekly spend for each campaign type equals the number of days the campaign runs in week t times its daily cost. The retailer used the same €.70 daily cost for each campaign type. This cost includes two main elements: copy and creative design, and the opportunity cost of house ads appearing on the retailer's website; website space is precious, and using it for one activity precludes using it for another. Marketing spend is not contingent on purchase because, as noted previously, purchase-related costs such as discounts and free shipping are included in order size OS_{it}.

Overall, the cost structure means that our decision variables are the number of days each campaign runs in week t, with a cost of €.70 per day. The sum of costs across the four campaigns equals total marketing spend.[12]

The optimization involves both immediate and long-term considerations. First, as shown in Figure 1, marketing directly increases purchases and returns. Thus, it immediately generates a benefit (more sales) and either an additional benefit (if it directly decreases returns) or a cost (if it directly increases returns). Second, dynamic effects operate across periods. Returns in period t directly determine purchase and returns in future periods (through either the virtuous or vicious cycles), and purchases in period t also determine purchases and returns in future periods.

How this plays out in optimal levels of marketing depends on the estimated statistical model parameters. One can nevertheless see the need for dynamic optimization to resolve short- and long-term trade-offs. For example, marketing may have a strongly positive impact in the current period if it increases purchases and does not have a positive direct effect on returns. However, more purchases lead to more returns through Equation 3, and those returns may increase purchases in the future due to the virtuous cycle. At the same time, the vicious cycle operates: Current returns increase both ReturnStock and returns in the future.

The optimization must determine weekly marketing spend that accounts for both virtuous and vicious cycles. Our model quantifies these relationships, but this is a dynamic optimization, not the sort of back-of-the-envelope calculation one might make. We test whether dynamic optimization improves over myopic optimization.

Customer-level purchases and returns are probabilistic, so we require a stochastic dynamic optimization. Because the firm used a one-quarter time horizon when specifying its marketing, we employ a finite horizon (Khan, Lewis, and Singh 2009). We use the generalized reduced gradient algorithm implemented in Frontline Solver, a nonlinear optimization algorithm. When it converges to a solution, it means that there is no other set of values for the decision variable close to the ones identified that yield higher profits over our time horizon. However, there may be better solutions far away from the one identified. We address the possibility that the algorithm identified a local optimum using multiple starting values (see Web Appendix B for more details on optimization specification).

As previously noted, we use myopic optimization as a benchmark to compare dynamic versus myopic (see Neslin et al. 2009, pp. 728–29). Our myopic optimization uses a simple rule: Find the optimal level of marketing activity each week without considering its implications for future marketing activities, which means considering only the current marketing efforts at time t. As Neslin et al. (2009) report, and as is sometimes the case, myopic optimization, a form of "greedy algorithm," can do quite well, especially if the effect sizes of dynamics are not very large.

Results

Estimates of the Purchase/Returns Model

ReturnStock (Equation 5) captures the magnitude of each customer's memory of past returns. Its coefficients in the statistical model represent its association with purchases and returns, that is, the vicious and virtuous cycles. We performed a grid search to estimate λ, allowing the value to range from 0 to 1 in increments of .1 (Dinner, Van Heerde, and Neslin 2014). Our first pass suggested we explore the area above .7 more granularly. Figure 5 shows the results, suggesting a peak between .9 and .95. We then narrowed the range and conducted another grid search using increments of .01; the figure shows a clear peak at $\lambda = .93$. This suggests using .93 as the estimate of λ (see Fader, Lattin, and Little 1992). Such a large value suggests that consumers remember return experiences for a long time,[13] and the coefficients for ReturnStock measure how this memory affects returns and purchases.

Table 4 presents the estimates of the proposed statistical model, Equations 1–4 (Model 1). These estimates confirm that returns are associated with future sales, as ReturnStock_{it} has a significant and positive impact on both purchase incidence ($\gamma^P = .226$, $p = .000$) and order size ($\gamma^{OS} = .118$, $p = .000$),

[12] When optimizing, we set constraints for marketing decision variables based on the range of data in Table 2 and practical considerations.

[13] Calculating Equation 5 over time shows that with $\lambda = .93$, 90% of the impact of returns occurs within 8.5 months, which seems reasonable given the relatively infrequent interpurchase time for fashion clothing.

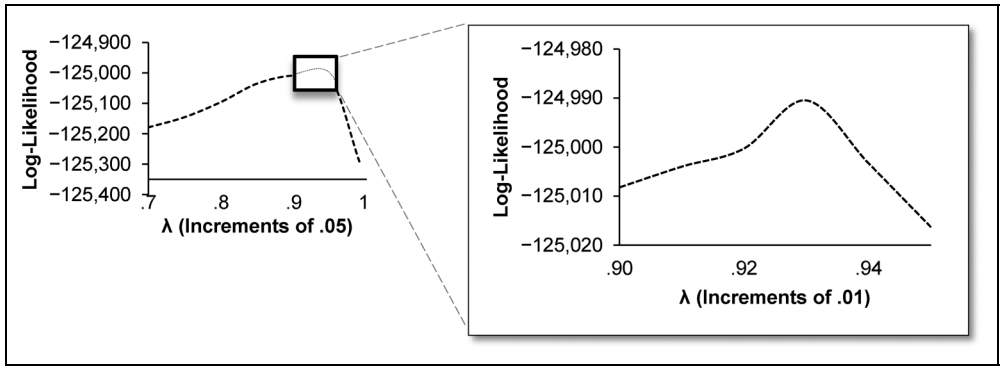


Figure 5. Estimating ReturnStock_{it} Carryover: Model Fit as a Function of λ.

Table 4. Model 1 (M1): Full Product Return Dynamics.

Macro Category	Variable	Purchase Incidence (P_{it}) Coef. (SE)	Order Size (OS_{it}) Coef. (SE)	Return Incidence (R_{it}) Coef. (SE)	Return Size (RS_{it}) Coef. (SE)
Marketing: non-brand-building activity	Marketing discounts (β_{MKD})	.010	.006	−.024	−.012
		(.004)	(.005)	(.012)	(.008)
	Free shipping (β_{FS})	.143	.110	.092	.028
		(.019)	(.023)	(.048)	(.028)
Marketing: brand-building activity	Retailer advertising (β_{RET})	.014	.006	−.002	.000
		(.002)	(.002)	(.005)	(.003)
	Brand advertising (β_{BRD})	.102	−.030	.013	.007
		(.028)	(.032)	(.066)	(.045)
Lagged marketing: non-brand-building activity	Marketing discounts lag ($\beta_{MKD.LAG}$)	.016	.017	.012	−.001
		(.004)	(.005)	(.012)	(.008)
	Free-shipping lag ($\beta_{FS.LAG}$)	−.022	−.021	.001	.032
		(.020)	(.025)	(.053)	(.028)
Lagged marketing: brand-building activity	Retailer advertising lag ($\beta_{RET.LAG}$)	−.005	−.004	.005	.001
		(.002)	(.002)	(.005)	(.003)
	Brand advertising lag ($\beta_{BRD.LAG}$)	−.064	−.053	.031	−.098
		(.029)	(.033)	(.077)	(.067)
Product return dynamics	Return stock (γ)	.226	.118	.249	.090
		(.007)	(.015)	(.020)	(.009)
State dependence	Purchase incidence lag (ι^{P})	.320			
		(.014)			
	Order size lag(ι^{OS})		.082		
			(.004)		
	Return size lag (ι^{RS})				.737
					(.025)
Recursive component	Order size (τ)			.325	.042
				(.061)	(.004)

Notes: Number of observations = 252,148, LL = −124,990.48, Wald χ^2 = 16,513.50, $p < .001$. The order size and return size variables are expressed as natural logarithms. The κ^{P}, κ^{R}, κ^{OS}, and κ^{RS} parameters for control variables and ν^{P}, ν^{R}, ν^{OS}, and ν^{RS} parameters for time trend variables are not reported. Complete results are available in Web Appendix C, Table W.C1.

capturing the virtuous cycle. The table also reveals a vicious cycle: ReturnStock_{it} positively impacts return incidence (γ^{R} = .249, p = .000) and return size (γ^{RS} = .090, p = .000). Returns increase future sales but generate more future returns as well.

Both non-brand-building marketing (β^{P}_{MKD} = .010, p = .000, β^{P}_{FS} = .143, p = .025) and brand-building marketing (β^{P}_{RET} = .014, p = .000, β^{P}_{BRD} = .102, p = .000) are positively and significantly associated with purchase. Free shipping and

retailer advertising also are associated with order size ($\beta^{os}_{FS} = .110$, $p = .000$, $\beta^{os}_{RET} = .006$, $p = .003$). Lagged marketing discounts are associated with purchase incidence and order size ($\beta^{P}_{MKD.LAG} = .016$, $p = .000$, $\beta^{os}_{MKD.LAG} = .017$, $p = .002$), which may reflect consumers waiting for deeper discounts.[14] Negative lags could reflect purchase acceleration, as purchases move forward in time. Free-shipping promotion is associated with purchase and order size and is the only marketing associated with return incidence ($\beta^{R}_{FS} = .092, p = .053$), supporting the power of these promotions (Shehu, Papies, and Neslin 2020).

We further test for dynamic effects of product returns by comparing the proposed Model 1 (M1; Equations 1–4) with two benchmarks: Model 2 (M2) limits return dynamics by imposing $\lambda = 0$ when computing $ReturnStock_{it}$, and Model 3 (M3) eliminates return dynamics by excluding $ReturnStock_{it}$ and RS_{it-1} from Equations 1–4. The results, presented in Web Appendix C, Table W.C2, confirm that the proposed model (M1) is superior to both M2 and M3 in terms of the Akaike information criterion and the Bayesian information criterion. This supports a dynamic relationship between product returns and sales.

In summary, the statistical model reveals the following: (1) a virtuous cycle, reflected in a positive association between returns and future purchases; (2) a vicious cycle, reflected in a positive association between returns and future returns; (3) a positive relationship between marketing and purchase; (4) a positive relationship between one marketing activity—free shipping—and return incidence; and (5) improved model fit relative to models that exclude return dynamics.

Determining how to free ride on the positive association of returns with purchase is a complex task given its dynamic associations with purchase incidence, order size, return incidence, and return size. Doing so requires marketing to capitalize on the virtuous cycle, which may entail adjustments to marketing spend, the scheduling of marketing activities, and the allocation of spend across marketing efforts. We present results along these three dimensions.

Free-Riding Product Returns: Optimal Marketing Spend

Table 5 reports that dynamic optimization increases profits by 46.56% compared with myopic optimization,[15] while decreasing both marketing spend (39.87% lower) and return costs (33.18% fewer returns at close to the same monetary value per return).

The table reports results based on the dynamic statistical model M1. However, this may not be necessary. Perhaps limiting or eliminating return dynamics—M2 and M3—would generate similar profits and therefore suggest that M1's complex modeling of dynamics is not required.

Table 5. The Full Dynamic Model (M1) Optimized Either Dynamically or Myopically: Average Sales, Returns, Profits, and Marketing Costs per Customer per Week.

	(M1/Myopic) (€)	(M1/Dynamic) (€)	% Difference
Sales	126.81	139.77	10.22
Marketing spend	15.08	9.07	−39.87
Returns	.40	.27	−33.18
Returns monetary value	15.66	15.85	1.23
Profits	26.17	38.36	46.56

Notes: Figures represent averages calculated across all customers and over Weeks 4–25.

To investigate further, we compute the optimal marketing policy dictated by M1, M2, and M3 under both dynamic and myopic optimization. This yields six policies: M1/Myopic, M1/Dynamic, M2/Myopic, and so on. M1 is treated as the "correct" model because it fits the data best. We therefore substitute the marketing plan prescribed by each of the six policies into M1 to compute profits. The objective is to quantify how much one would lose by using the "wrong" statistical model/optimization combination.

Figure 6 shows profits for the six policies. The two bars on the far right are from Table 5: Dynamic optimization of M1 generates more profit than myopic optimization. The figure adds two insights. First, dynamic optimization outperforms myopic whenever the statistical model includes dynamics: M1/Dynamic exceeds M1/Myopic, M2/Dynamic exceeds M2/Myopic, but M3/Dynamic does not exceed M3/Myopic. Because M3 omits all return dynamics, dynamic optimization has no leverage. Second, dynamic optimization generates more profit the closer the statistical model gets to the fully dynamic M1 (the black bars in the figure), whereas myopic optimization does not (the gray bars). Because myopic optimization does not consider dynamics, it cannot exploit a statistical model that includes them. Overall, the figure shows that it is important to include dynamics in both the statistical model and the optimization.

Free-Riding Product Returns: Optimal Marketing Scheduling

Figure 7 presents the marketing spend and returns patterns for M1/Dynamic,[16] showing a pulsing pattern in marketing and a pulsing pattern in ReturnStock one week out of phase with the spend pattern. A marketing pulse increases purchases. Some are returned, which increases ReturnStock in the

[14] In fact, the firm typically increases discount rates progressively (e.g., from 60% in the first week to 80% in the second week). Consumers adjust their behavior for this (Gonul and Srinivasan 1996).

[15] The early periods of a dynamic optimization are often transitory. For our optimization, stability began in the fifth period (t_4). We therefore state profits and comparisons from t_4 through t_{25}, which we call the "evaluation period."

[16] Profit calculations are based on t_4 through t_{25}. Also note that t_4 begins with marketing spend not hitting 0 (Figure 7) and myopic showing a "bump" (Figure 8), as both optimizations used MKD in t_4.

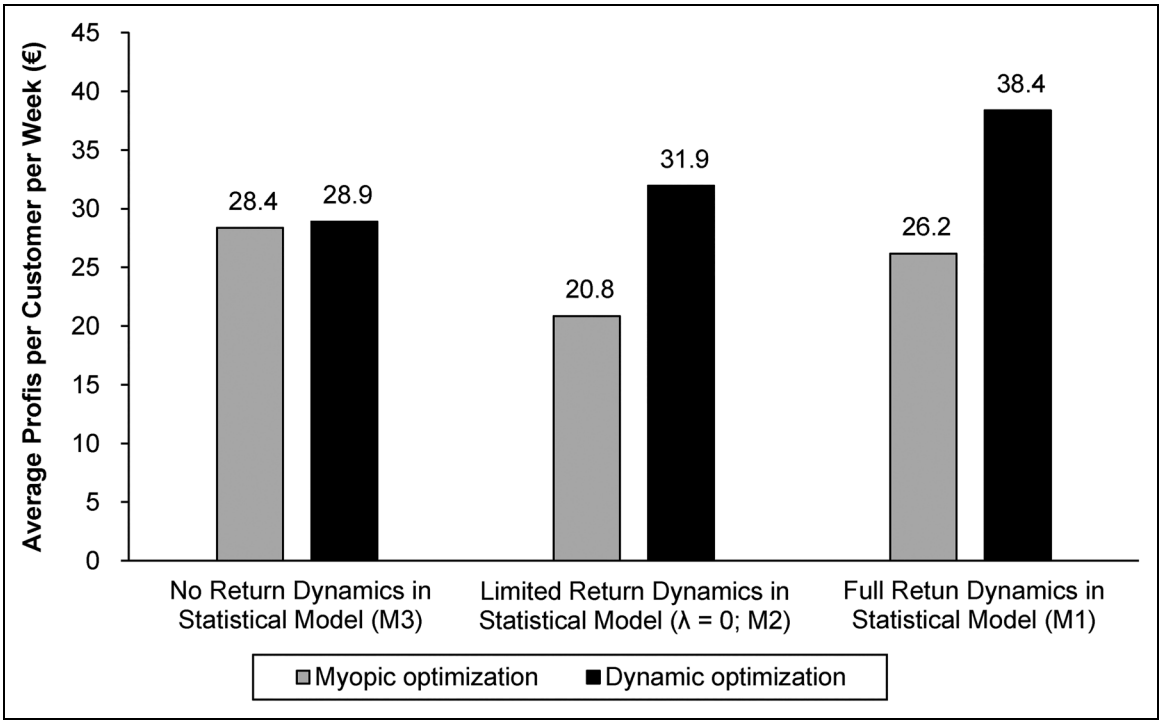


Figure 6. Optimal Profits Generated by Each Descriptive Model/Optimization Combination.
Notes: M1 (full model with $ReturnStock_{it}$): the full proposed model described in Equations 1–6. M2 (limits return dynamics): uses $\lambda = 0$ to compute $ReturnStock_{it}$, thereby limiting return dynamics. M3 (no return dynamics): completely eliminates return dynamics by excluding $ReturnStock_{it}$ and RS_{it-1} from Equations 1–6.

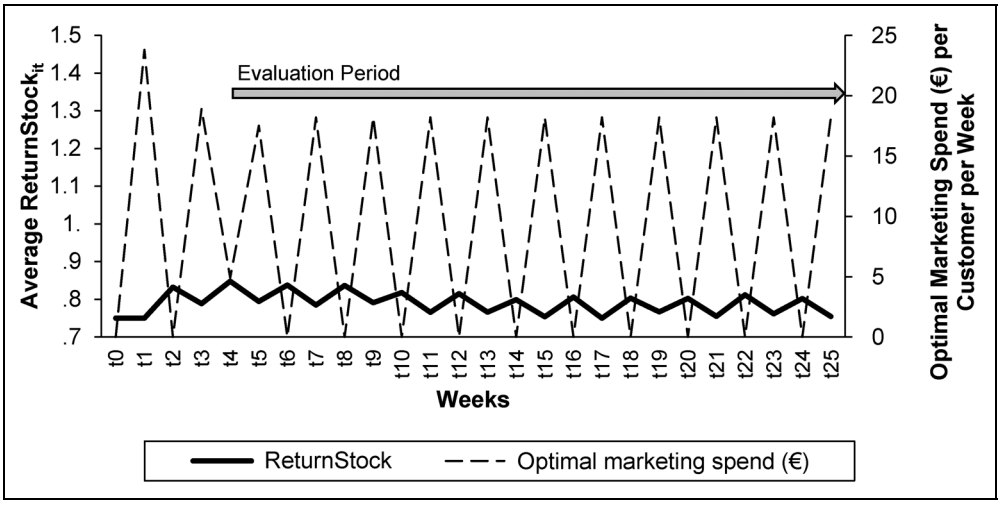


Figure 7. Dynamic Optimization: Average $ReturnStock_{it}$ and Marketing Spend.
Notes: The early periods of dynamic optimization are often transitory as the model approaches a stable solution. The progression of the key variable ReturnStock suggests that for our optimization, stability began in the fifth period (t_4). We therefore report all profits and profit comparisons from t_4 through t_{25}—the "evaluation period"—in Figures 7 and 8.

following week. We can free ride on this increase because larger ReturnStock is virtuous: It stimulates purchase, allowing marketing spend to be reduced when ReturnStock increases. The crucial insight is that firms can increase profits by using marketing to exploit the positive effects of returns—the virtuous cycle.

Figure 8 presents the myopic optimization of M1. It shows a hint of pulsing but not nearly to the extent observed under dynamic optimization. Myopic optimization does not capitalize on the virtuous cycle because it does not consider the future benefits of current actions.

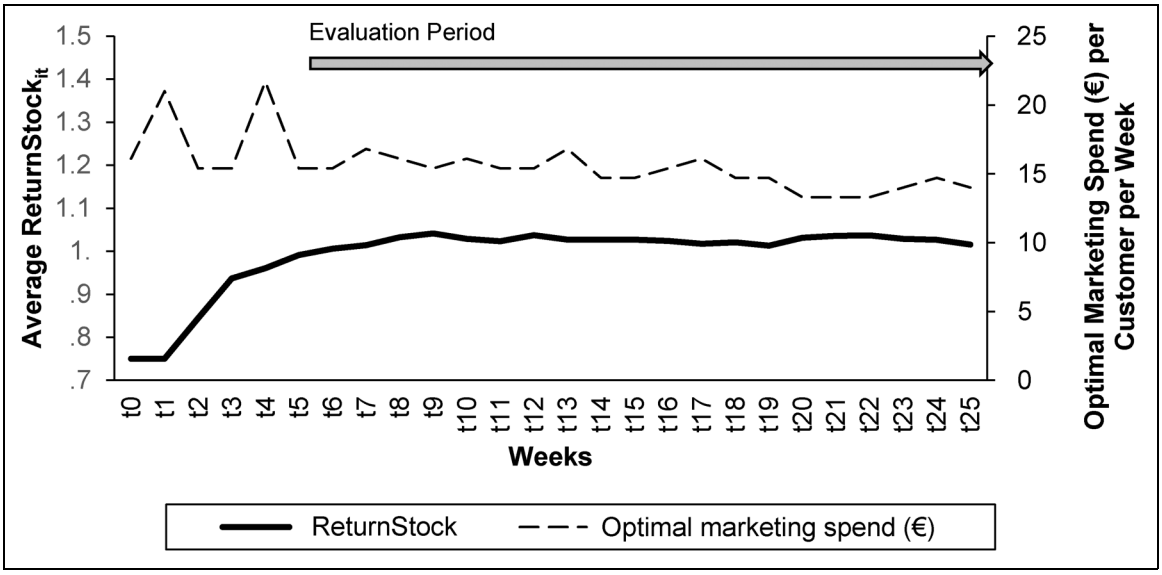


Figure 8. Myopic Optimization: Average ReturnStock$_{it}$ and Marketing Spend.

Table 6. Marketing Spend Allocation: Full Dynamic Model (M1) Optimized Either Dynamically or Myopically.

Type of Marketing Activity		Optimal Average Marketing Spend (€) per Customer per Week		Allocation Proportion (%)		Comparison in Allocation Proportion (%) Dynamic vs. Myopic Optimization
		Myopic Optimization	Dynamic Optimization	Myopic Optimization	Dynamic Optimization	
Non-brand-building	Marketing discounts	.22	.22	1.45	2.40	.94
	Free shipping	4.33	2.13	28.22	22.95	−5.27
Brand-building	Retailer advertising	5.06	3.50	32.99	37.67	4.68
	Brand advertising	5.73	3.44	37.34	36.99	−.36
Total		15.34	9.29	100	100.00	100.00

Free-Riding Product Returns: Optimal Marketing Allocation

The success of free riding in Figure 6 entails scheduling—a pulsing strategy for M1/Dynamic. Table 6 shows the role of reallocating spend across marketing activities. Most notably, dynamic optimization allocates 22.95% of the marketing budget to free shipping and 37.67% to retailer advertising. In contrast, myopic optimization allocates 28.22% to free shipping and 32.99% to retailer advertising.

The parameter estimates in Table 4 suggest why this happens. Free shipping is strongly associated with purchase incidence and order size, initiating the virtuous cycle. However, it also stimulates returns. Some of these returns contribute to the virtuous cycle, but they also stimulate future returns, the vicious cycle. The myopic optimization does not account for this and therefore allocates more of its budget to free shipping. Dynamic optimization, in contrast, recognizes the vicious cycle and pulls back from free shipping.

Dynamic optimization also realizes that retailer advertising positively relates to both purchase incidence and order size, the only marketing other than free shipping that does this. However, retailer advertising does not directly increase returns, and therefore more funds can be allocated to it to generate the virtuous cycle without generating too many returns. Recognizing this requires forward-looking optimization, which myopic optimization does not perform.

Again, we see that dynamic optimization works because it incorporates future effects into its calculations. It therefore allocates funds among marketing activities to leverage the virtuous cycle without creating too much of the vicious cycle.

Quantifying Free-Riding Opportunities Across Geographical Regions and Return Policies

The focal firm operated in several countries, and its return policy varied across countries due to country-specific

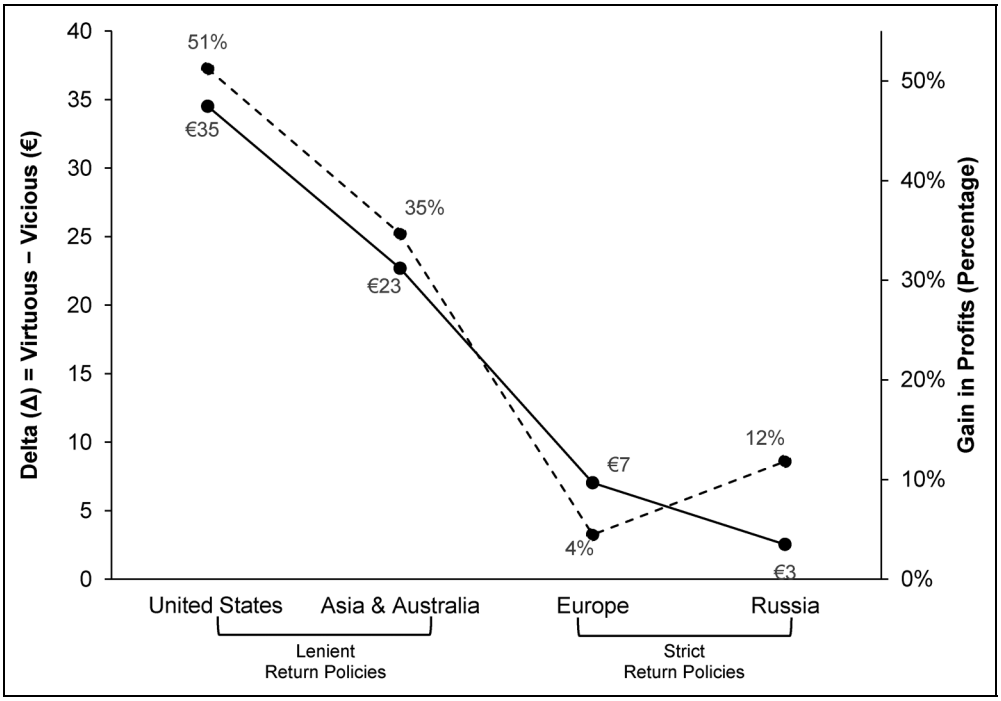


Figure 9. Quantifying the "Free-Riding" Opportunity.
Notes: Europe includes only countries with stricter return regulations. The left y-axis is the excess of the virtuous cycle over the vicious cycle: Delta = (Virtuous Cycle − Vicious Cycle) = $(P_{it} \times OS_{it}) - (R_{it} \times RS_{it})$. The right y-axis reports the gain in profits: $\frac{\text{Profits (M1 Dynamic)} - \text{Profits (M1 Myopic)}}{\text{Profits (M1 Myopic)}} \times 100$.

competition and culture. The United States and Asia/Australia offered lenient policies entailing free returns and up to 100 days between purchase and a permissible return. Russia and European countries such as Denmark, Greece, and the Netherlands had strict return policies, with payment required for returns and short return windows (e.g., 15 days). This variation provides an opportunity to quantify free riding by examining the impact of lenient versus strict return policies.

Our analysis consists of the following steps for each country. First, we estimate the fully dynamic statistical model, M1. We then calculate the virtuous and vicious cycles using M1 and compute "delta," defined as the difference between them. Next, we optimize marketing using M1 with both dynamic and myopic optimization; the difference between the two indicates free riding. Last, we relate this difference to delta.

Bower and Maxham (2012) and Wood (2001) suggest that lenient return policies increase the virtuous cycle, and thus delta should be highly positive for countries with lenient policies. This suggests that lenient policies are associated with more free riding through the following logical chain: large delta (virtuous minus vicious cycles) ≥ large difference in dynamic minus myopic optimization ≥ free riding. We investigate by following the steps outlined previously.

Step 1: We estimate M1 for four geographical regions: two representing lenient policies (United States, Asia and Australia) and two representing strict policies (European countries listed in Table W.D1 and Russia). The results are detailed in Web Appendix D.[17]

Step 2: We calculate revenues (purchase incidence × order size) when ReturnStock_{it} is at its mean plus one standard deviation, minus revenues when ReturnStock_{it} is at its mean. This is the virtuous cycle. The vicious cycle is return costs (return incidence × return size) when ReturnStock_{it} is at its mean plus one standard deviation, minus costs when ReturnStock_{it} is at its mean. See Web Appendix E for details.

Step 3: The left y-axis in Figure 9 plots delta, the difference between virtuous and vicious cycles for each region. For example, delta for the United States is €35.

Step 4: We maximize profits using dynamic and myopic optimization—the difference between the two represents free riding. We graph this difference on the right y-axis in Figure 9. The United States gains 51% in profits using dynamic rather than myopic optimization.

Step 5: Figure 9 shows the correspondence between delta, the excess of virtuous over vicious cycles, and the profit benefit achieved by dynamically taking advantage of this

[17] As marketing activity varies by country, there is a potential concern for endogeneity in aggregate, cross-country analysis. By running separate models and optimizations per geographical regions, we are accounting for country-specific variations in marketing efforts. We thank an anonymous reviewer for this suggestion.

excess. The free-riding benefit mirrors the difference between virtuous and vicious cycles.[18]

Figure 9 shows that lenient policies are associated with larger gaps between virtuous and vicious cycles and larger profit gains from dynamic optimization that exploits free riding. In other words, delta ≥ gain from dynamic optimization ≥ more free riding. To further investigate, we created a regional version of Figure 7 (Web Appendix D, Figure W.D1) and found a pulsing pattern in ReturnStock for lenient policy regions and none in strict policy regions.

These results suggest that a large, positive difference between virtuous and vicious cycles creates free-riding opportunities. This difference is related to the return policy: Lenient policies amplify the difference, enabling firms to capitalize on returns—to free ride on them—to increase profits. Conversely, strict policies shrink this advantage, limiting the potential to fully leverage returns. The patterns in Figure 9 are striking. Because the evidence is based on a "natural experiment" due to country differences, we cannot unequivocally claim causality. Nevertheless, the findings encourage future randomized field tests.

Discussion and Implications

Summary

The conventional wisdom is that returns are "bad"—a vicious cycle—but the intriguing finding is that current returns are associated with higher sales in the long run—a virtuous cycle. The thesis of this article is that if marketing is specified optimally, the firm can create and translate the virtuous cycle into higher profits, that is, can free ride on returns.

We investigated whether combining a dynamic descriptive model with dynamic optimization can increase profits through free riding. The answer is yes. Optimal marketing in the presence of a virtuous cycle follows a pulsing pattern: return stock, representing consumers' memory of returns, builds during low-marketing periods, amplifying the virtuous cycle. Dynamic optimization generates this pulsing behavior, whereas myopic does not, suggesting that both a dynamic statistical model and dynamic optimization are needed to achieve free riding. Moreover, dynamic optimization refrains from overinvesting in free-shipping promotions because these generate excessive returns—leading to a larger vicious cycle. Finally, lenient return policies are associated with larger differences between virtuous and vicious cycles and with larger profit differences between dynamic and myopic optimization, suggesting that lenient policies produce more free riding.

Our work fits within the DSS paradigm. A DSS incorporates four ingredients (Little 1979; Lilien, Kotler, and Moorthy 1992): a model, statistics, data, and optimization. The model conceptualizes the key phenomena under study—in our case, the virtuous and vicious cycles. The statistical component is the quantification of the model through the four equations we developed and estimated, and the data consist of the 1,500 customers we analyzed. Optimization is the stochastic dynamic program we employed to generate an optimal free-riding strategy.

The DSS produces both optimal marketing policies and managerial insights, as shown in Tables 5 and 6. The insights include the following: First, marketers may be able to free ride on the virtuous cycle if they develop marketing plans that generate a large virtuous cycle without generating too large a vicious cycle. Second, the goal is not zero returns but a marketing strategy that exploits the virtuous cycle while accounting for the vicious cycle. Third, the right marketing policy may require reallocating spend among marketing activities rather than simply decreasing spending. Fourth, the opportunity to free ride depends on the firm's return policy. Strict policies provide less opportunity to free ride—they do not generate a strong enough virtuous cycle. Fifth, the build-up and decay of consumer memory quantified by return stock generate virtuous and vicious cycles. Finally, free riding requires dynamic statistical and optimization models, which generate superior profits relative to when the empirical model, optimization, or both are not dynamic. Free riding is a dynamic strategy—it requires dynamic measurement and optimization.

Our work contributes in three ways. First, it integrates dynamic optimization with empirical research that identified virtuous and vicious cycles. Second, consistent with the DSS perspective, it provides insights, as noted previously. For example, our finding regarding the value of lenient return policies is provocative because many managers naturally shy away from making it too easy to return. Third, also consistent with the DSS paradigm, we show how to implement the free-riding strategy, particularly the requirement to include dynamics in both the statistical model and optimization.

Research Implications

Product returns must be included in transaction data models of consumer purchase, as they are associated with an immediate and long-term impact on sales. The long-term impact also requires that customer-targeted marketing maximize customer lifetime value. While there is strong support for the virtuous cycle, it is less obvious than the vicious cycle; therefore, future research should measure it in different contexts (e.g., multichannel retailing environments). Additionally, the exact mechanism that creates the virtuous cycle must be ironed out. Bower and Maxham (2012) and Wood (2001) make a promising start regarding perceived fairness. However, several possible mechanisms remain and need further investigation. Additional insights into these mechanisms are provided in Web Appendix F.

Dynamic statistical modeling coupled with dynamic optimization represents the most effective configuration, and our results underscore how acute the differences are when one

[18] We perform a sensitivity analysis where we compare our best model (M1/Dynamic) with another benchmark (M3/Dynamic). The pattern shown in Figure 9 remains consistent across this comparison.

Table 7. Marketing Spend Allocation: Actual Versus Optimal: Average per Customer per Week.

Type of Marketing Activity		Actual[a]		Optimal	
		Spend (€)	Proportion (%)	Spend (€)	Proportion (%)
Non-brand-building	Marketing discounts	.13	11.70	.22	2.40
	Free shipping	.11	9.85	2.13	22.95
Brand-building	Retailer advertising	.43	39.82	3.50	37.67
	Brand advertising	.42	38.63	3.44	36.99
Total		1.09	100.00	9.29	100.00

[a]To ensure consistency with the average optimal marketing, we calculated the average amount spent on various marketing activities during Weeks 4–25 in the observed sample.

relaxes either the statistical or the optimization dynamics. In addition, the role of competition warrants further investigation. Does competition learn that the focal firm is pulsing and hence market out of phase in the belief that it can override the firm's free riding? What is the marketing and profit equilibrium when multiple firms can free ride?

Future research should also consider customer heterogeneity. Certain customers may be susceptible to the virtuous cycle, whereas others may be susceptible to the more vicious cycle. Finally, we compiled our analysis sample to focus on every-year buyers. One of the benefits of this approach was mitigating survivor bias by excluding "churners." However, controlling churn is a significant issue; future work should therefore examine the relationship between vicious and virtuous cycles and customer churn.

Finally, our analysis, and much of the empirical work on returns, employs observational data. To explore this further, we conducted two investigations detailed in Web Appendix F. The first was propensity scoring matching (PSM). The treated group consisted of customers who purchased in a given month and returned a product, and the control group consisted of those who purchased in that month but did not make a return. We repeated the analysis across several different time periods using different PSM specifications. Our findings consistently revealed a positive average treatment effect, comparable in magnitude to what one would infer from our estimated model in Table 4.

We then conducted a preregistered, randomized online experiment where we manipulated the returns policy—lenient versus strict. As expected, the lenient policy generated more returns, thereby exogenously creating high- and low-returns groups. We found that the lenient group's self-reported purchase intentions were larger than those of the strict group, supporting the virtuous cycle. In addition, the experiment generated insights into the mechanisms underlying the virtuous cycle. Mediation analysis suggested that perceived fairness, relationship quality, and lower perceived risk of purchasing the wrong product were key mechanisms translating the lenient policy into larger future purchase intentions.

These analyses support our findings of a positive relationship between returns and purchases—a virtuous cycle. However, they do not prove causality in the real world. The PSM remains subject to unobserved causal factors, and the online experiment is subject to measurement specifics and questions of external validity. For this reason, we strongly recommend field experiments to provide more convincing evidence of virtuous and vicious cycles.

Managerial Implications

The good news for managers is that returns are not the enemy; rather, the enemy is the opportunity loss from not managing them effectively. Unfortunately, there is no simple prescription. We can say that marketing is the fulcrum, but we cannot generalize how much to spend on which types of marketing. The free-riding strategy makes sense, but it requires a carefully optimized marketing schedule. With so much at stake, we suggest that managers replicate our statistical model, combine it with dynamic optimization, and test its implications. Our methods, while nontrivial, are within reach of well-trained data scientists.

Our DSS viewpoint guides managers on how to use the raw numbers the firm collects from its customers. The mantra is model, statistics, data, and optimization, where the model is "a preconceived idea of how the world works and therefore of what is interesting and worthwhile in the data" (Little 1979, p. 10). Our Figure 1 framework drives our DSS by pointing to the key phenomena—virtuous and vicious cycles—that managers must balance. Statistics involves specifying and estimating a set of equations (Equations 1–6) that follow from the framework, and optimization refers to any effort to improve the firm's current strategy. The statistical model provides the parameter values required to run the optimization and on its own yields the important insights noted previously. While it is an important component of the DSS, the statistical model is based on observational data. Hence, it provides valuable descriptive insights that support decision-making, but we cannot guarantee that the results are causal.

The goal of a DSS is to improve managerial decisions, which suggests comparing the "optimal" marketing policy with that currently employed by the focal firm. Table 7 reports this comparison. The "Optimal" columns are from the "Dynamic Optimization" allocation columns in Table 6, and the "Actual" columns are drawn from records the retailer shared with us. The retailer appeared to be underspending on marketing, perhaps unaware of the power of marketing. Interestingly, however, it allocated funds roughly consistent with our DSS-generated recommendations: 78% to brand-

building efforts such as retailer brand and manufacturer brand advertising and 22% to more promotional activities such as price discounts and free shipping. We would recommend reallocating some funds from discounting to free shipping, though not to overdo it. Such insights are the hallmark of the DSS paradigm.

By comparing results at the country/region level, we show that a strict returns policy generates less of a virtuous cycle and thus less free riding. In countries where the firm adopts a lenient policy, the difference between virtuous and vicious effects is larger, creating the opportunity to free ride on returns. The gains from pursuing this opportunity are substantial (Figure 9). The key takeaway is a point of view: Rather than treating returns as something to be suppressed through strict policies that decrease returns but alienate consumers, the firm should consider lenient policies that create virtuous cycles and increase profits in the long run.

In summary, we build on previous findings in marketing to show that returns can be managed to increase profits. The lever is marketing; the key phenomena are the dynamics of returns-induced virtuous and vicious cycles. Incorporating dynamics into both statistical models and optimization yields the most effective means for marketing to exploit the virtuous cycle and increase profits. Indeed, we hope that our work stimulates researchers to continue the virtuous cycle of building on previous work to replicate, validate, and extend our results.

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Data Availability

An alternative disclosure plan was approved for this article.

The data that support the findings of this article are not available due to a nondisclosure agreement with the partnering firm. However, replication materials (including a synthetic dataset that mimics the structure of the original data and all analysis code) have been uploaded to the *JM* Dataverse in restricted form and are available from the authors on reasonable request, subject to the terms of the NDA.

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